THE MASTERMIND COACHING CLASSES

Revision Notes

Class 9 · Science

NCERT Curiosity — Chapters 8 to 13 · CBSE / RBSE Syllabus

HOW TO USE THESE NOTES

Each chapter follows the textbook's own order of sections, so you can read straight through. Bold words are keywords you must be able to define. Tables compare and classify; shaded boxes hold definitions, formulas and key facts. The navy Key Takeaways box at the end of every chapter is your last-two-minutes revision.

CH.	CHAPTER TITLE	CORE THEME
8	Sound Waves: Characteristics and Applications	Production, propagation, characteristics, echo, ultrasound
9	Cell: The Building Block of Life	Cell structure, organelles, osmosis, cell division, cell theory
10	Tissues in Action	Plant and animal tissues, joints and the skeletal system
11	Patterns in Life: Diversity and Classification	Biodiversity, five kingdoms, plant and animal groups, binomial naming
12	Reproduction: How Life Continues	Asexual and sexual reproduction in plants, animals and humans
13	Earth as a System: Energy, Matter and Life	Spheres, solar radiation, winds, biogeochemical cycles, human impact

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CHAPTER 08 · PHYSICS

Sound Waves: Characteristics and Applications

Sound is energy that reaches us as a wave. This chapter explains how sound is made by vibrations, how it travels through a medium as compressions and rarefactions, what quantities describe a wave (wavelength, frequency, time period, amplitude, speed), how we perceive it (pitch, loudness), and how reflection, echo and ultrasound are put to use.

Big Questions

- How is sound produced and how does it reach our ears?
- Why can astronauts on a spacewalk not talk directly to each other?
- How do bats find prey in the dark?

8.1 Production of Sound

Every sound comes from something that is **vibrating**. A **vibration** is a periodic to-and-fro motion (oscillation) of an object. The object that makes the sound is the **source** of sound.

Activity — rubber band on an open box
Pluck a stretched rubber band: you hear a sound as long as the band is visibly vibrating, and the sound stops when the vibration stops. Changing the tension changes the sound. A band held between two fingers also sounds, but much softer, because the box helps the sound to be louder.

Activity — tuning fork
Strike a tuning fork on a soft rubber pad (never on a hard surface) and hold it near the ear: sound is heard. Touch a vibrating prong to water: waves splash and ripple, proving that the prongs vibrate.

Source of sound	What vibrates	Example
Stretched string	The string itself	Sitar, veena, rubber band
Membrane (stretched skin)	The membrane	Tabla, drum
Air column	Air inside a hollow pipe	Bansuri (flute)
Metal bar / prongs	The metal	Tuning fork, taal (cymbals)
Vocal cords (humans)	Stretched muscular flaps in the voice box (larynx)	Speech, singing; tongue, lips, mouth and nasal cavity shape it into words
Body parts rubbed together	Wings or legs	Grasshopper, cricket

- In most musical instruments more than one part vibrates.
- A **tuning fork** is a U-shaped steel or aluminium bar with a stem; the two sides of the U are the **prongs (tines)**. It gives an almost single-frequency sound.
- Kongthong (Meghalaya), the 'Whistling Village', gives every person a personal tune name (*Jingrwai Iawbei*), composed by the mother at birth.

8.2 Propagation of Sound

Sound travels from the source to our ears through a **medium** — the material through which it propagates. A **vacuum** is a space with no matter in it.

Activity	Observation	Conclusion
Ear on desk while a friend taps it	Sound is heard through the desk	Sound travels through solids
Tap two spoons under water	Tapping is heard through water and air	Sound travels through liquids
Normal listening in a room	Sound is heard	Sound travels through gases (air)
Electric bell in a bell jar, air pumped out	Sound fades to almost nothing although the bell is seen ringing; it returns when air is let in	Sound cannot travel in vacuum — it needs a medium

IMPORTANT

Outer space is a near vacuum, so astronauts on a spacewalk cannot hear each other or the clank of metal directly. They talk through radio devices fitted into their spacesuits. Light, in contrast, does travel through vacuum, which is why sunlight reaches us.

8.3 Sound Waves

▸ The slinky analogy

Push and pull one end of a slinky: regions where the turns are crowded together and regions where they are spread out travel along its length. A mark made on one turn only **oscillates about its own position, parallel to the disturbance** — it does not travel along. Sound in air behaves in the same way.

▸ Compressions and rarefactions

Imagine a long tube of air with an oscillating piston at one end. Without oscillation the air has a uniform **average density**.

- When the piston moves forward it pushes nearby particles together: a region of **higher than average density** forms, called a **compression (C)**.
- When the piston moves back, a region of **lower than average density** forms, called a **rarefaction (R)**.
- Collisions between neighbouring particles pass each compression and rarefaction forward. A **series of alternating compressions and rarefactions** is a **sound wave**.
- The direction in which the wave moves is the **direction of propagation**.

IMPORTANT

In a sound wave the **disturbance (energy) travels, the particles do not**. Each particle only vibrates about its mean position. A small point source sends spherical waves in all directions.

▸ Types of waves

Term	Meaning	Example
Longitudinal wave	Particles vibrate **parallel** to the direction of propagation	Sound; longitudinal seismic waves (first to be recorded by seismographs)

Term	Meaning	Example
Transverse wave	Particles vibrate **perpendicular** to the direction of propagation	Light, waves on a string; some seismic waves
Mechanical wave	Needs a material medium	Sound (longitudinal), water waves
Non-mechanical wave	Needs no medium	Light (travels through vacuum)

Sudden loud sounds (firecrackers, thunder) arise when gases are heated rapidly and expand quickly, sending out a sudden pulse of density change. A **supersonic aircraft**, flying faster than sound, produces a loud **sonic boom**.

8.4 Energy of Sound Waves

Spread grains of rice over a stretched rubber or cellophane sheet on a wide container and make a loud sound near it (without touching): the sheet vibrates and the grains **jump**. Louder sound makes them jump higher. Hence **sound is a form of energy**, and a sound wave transfers energy from the source through the medium by vibrations and collisions of particles — not by moving the medium itself.

- Particles of a medium are never truly at rest (thermal motion); a passing sound wave only temporarily increases their vibration.
- A **microphone** turns sound into an electrical signal (sound makes a thin diaphragm vibrate). A **speaker** does the opposite: an electrical signal makes a cone or diaphragm vibrate and produce sound.

8.5 Graphical Representation of a Sound Wave

- At a given instant, plot **distance on the x-axis** and **density of the medium on the y-axis**; a dashed horizontal line marks the average density.
- Above the line is a **compression**; the highest point is the **crest** (maximum density).
- Below the line is a **rarefaction**; the lowest point is the **trough** (minimum density).
- A graph of density against **time** at one fixed position can also be drawn.

8.6 Characteristics of a Sound Wave

Quantity	Definition	Symbol	SI unit
Wavelength	Distance between two consecutive crests or two consecutive troughs	λ	metre (m)
Frequency	Number of density oscillations per unit time at a fixed point	ν	hertz (Hz) = s^{-1}
Time period	Time taken for one complete density oscillation at a point	T	second (s)
Amplitude	Maximum change in density (in a compression or rarefaction) from the average density	—	—
Intensity	Sound energy passing per unit time through a unit area perpendicular to the direction of propagation	—	W m^{-2}
Speed	Distance a crest (or trough) travels in unit time	v	m s^{-1}

FORMULA

$\nu = 1 / T$ and $v = \lambda \times \nu$ (speed = wavelength × frequency). A shorter time period means a higher frequency.

▸ Frequency and amplitude

- Everyday sounds are a mixture of frequencies; a tuning fork or a careful whistle gives an almost single frequency.
- Musical notes Sa, Re, Ga … each have a distinct frequency; **Sa has the lowest** of the octave and the others rise in order.
- Larger amplitude means more energy; a harder strike makes grains jump higher.
- **Intensity falls with distance** because the same energy spreads over a larger area. A sound of larger initial amplitude travels farther before it dies out.

▸ Speed of sound

Medium (at about 15 °C)	Approximate speed
Steel (solid)	5000 m s^{-1}
Water (liquid)	1500 m s^{-1}
Air (gas)	340 m s^{-1}

- Sound is **fastest in solids, slower in liquids, slowest in gases** (about 4–5 times faster in water and 15–20 times faster in solids than in air).
- In air the speed **rises with temperature and humidity**: about 331 m s^{-1} at 0 °C and 344 m s^{-1} at 22 °C.
- In most media the speed depends only on the medium, not on the source or the frequency. If the frequency changes, the **wavelength changes and the speed stays constant** (so music of different notes arrives together).
- Lightning is seen before thunder because light is vastly faster; distance to the strike = speed of sound × time delay (5 s → 1.7 km).

▸ Worked numericals (method)

Problem	Method	Answer
10 oscillations in 2 s	$\nu = 10/2$; $T = 1/\nu$	5 Hz and 0.2 s
Wavelength at 20 Hz and 20 kHz in air (344 m s^{-1})	$\lambda = v/\nu$	17.2 m and 1.72 cm
Wave in steel, λ = 50 m, v = 5000 m s^{-1}	$\nu = v/\lambda$; $T = 1/\nu$	100 Hz and 0.01 s
Clap, echo after 0.5 s (v = 340 m s^{-1})	d = vt/2 (to and fro)	85 m
Sonar returns after 0.90 s (v = 1530 m s^{-1})	d = v × (t/2)	688.5 m

8.7 Human Perception of Sound

Time period, wavelength, frequency, amplitude and speed are physical and measurable. **Pitch** and **loudness** describe how we *experience* sound and are subjective.

Perceived quality	Linked to	Details
Pitch	Frequency	High frequency → shrill, high pitch (whistle, siren); low frequency → deep, low pitch (thunder, aircraft rumble). The relation is not exactly mathematical.
Loudness	Amplitude	Larger amplitude is heard louder, and it falls as we move away. Intensity can be measured; loudness depends on the listener.
Timbre (quality)	Overtones	Makes the same note on a flute, ektara or tabla sound different; depends on shape, material and construction of the instrument.

- Voices differ because of frequency and how the throat, mouth and nasal cavity shape sound. In adolescence boys' vocal cords lengthen and thicken, vibrate less often, and the voice deepens.
- **Audible range: 20 Hz to 20,000 Hz (20 kHz)**; it varies by person and shrinks with age. Below 20 Hz is **infrasonic**, above 20 kHz is **ultrasonic**.
- Dogs, cats, bats and dolphins hear ultrasound; elephants hear infrasound.
- Loudness is measured in **decibels (dB)**: rustling leaves a few dB, conversation about 60 dB, firecrackers above 100 dB. A small rise in dB is a large rise in intensity.
- **Noise** is unwanted or harmful sound; long exposure to loud noise harms health, sleep and hearing. Hearing loss is tested with **audiograms** and helped by **hearing aids** (microphone, amplifier, speaker).

▸ Hearing and music (extra)

- Sound makes the **eardrum** vibrate; tiny bones amplify the vibrations; the **cochlea** converts them into electrical signals for the brain. Two ears let the brain find direction from the tiny time gap (under 1/1000 s).
- A **tone** has a single frequency. A **musical note** combines a lowest **fundamental** frequency with higher **overtones**. An **octave** is the interval in which one note has double the frequency of the other (200 Hz and 400 Hz).
- Sir C. V. Raman (India's first science Nobel, for the Raman Effect) also studied the acoustics of the tabla and mridangam; their central black patch, the **syaahi**, gives rich tonal control.

8.8 Reflection of Sound

Sound bounces off solids and liquids and obeys the **same laws of reflection as light**: the angle of incidence equals the angle of reflection, and the incident ray, reflected ray and normal lie in one plane.

▸ Echo

- An **echo** is the sound heard again after reflection from a distant hard surface (cliff, mountain, long corridor).
- The ear can separate two sounds only if they are **at least 0.1 s** apart. In 0.1 s sound covers 340 × 0.1 = 34 m to and fro, so the **minimum distance to the reflector is 17 m**.
- Hard, smooth surfaces give strong echoes; curtains and other soft surfaces absorb sound; rough surfaces scatter it, so no clear echo is heard. In a small room reflections return too soon to be separated.

▸ Reverberation

- Repeated reflections in a large hall make sound **persist after the source stops**. It occurs when reflections arrive less than 0.05 s apart.
- Too much causes garbled sound. Auditoriums are designed for suitable reverberation, using absorbing panels, upholstered seats and curtains.

- Gol Gumbaz at Bijapur has a whispering gallery in which a faint whisper is heard several times across the dome.

Feature	Echo	Reverberation
Time gap between sounds	At least 0.1 s	Less than 0.05 s
What we hear	Sound repeated separately	Original sound prolonged
Where	Open spaces, cliffs, long corridors	Large halls and auditoriums

8.9 Ultrasonic and Infrasonic Waves and Their Applications

Type	Frequency	Uses
Infrasonic	Below 20 Hz	Detecting earthquakes, volcanic eruptions and severe storms (they travel far through air and Earth)
Audible	20 Hz – 20 kHz	Hearing, speech, music
Ultrasonic	Above 20 kHz	Ultrasonography (imaging organs without surgery); breaking kidney stones; ultrasonic welding; cleaning delicate machine parts; finding defects inside metal blocks; locating objects

▸ Echolocation and sonar

- **Echolocation**: bats emit short bursts of ultrasound and use the echoes to sense the position of obstacles and prey in the dark. Dolphins, whales and some birds do the same.
- **Sonar (sound navigation and ranging)**: a ship sends ultrasound into water and analyses the reflected waves to find the distance, direction and speed of submarines, shipwrecks and the sea bed. Depth = speed × (time / 2).
- **Audio surveillance**: low-frequency hum of drones and aircraft can be detected by sensitive sound sensors.
- Frontier: sounds recorded on Mars; timing earthquake sound to measure sea temperature; mosquito buzz used to identify disease carriers; soil microbe crackles to study soil health.

8.10 Quick Exam Questions

Likely question	Model answer
How is sound produced?	By vibrating objects, e.g. a plucked string, a struck tuning fork or vocal cords.
Why can sound not travel in vacuum?	A sound wave is mechanical; it needs particles to pass on compressions and rarefactions. The bell-jar experiment shows the sound fading as air is removed.
Why is sound called a longitudinal wave?	The particles vibrate parallel to the direction in which the wave travels.
What is the difference between a compression and a rarefaction?	A compression is a region of higher density than average; a rarefaction is a region of lower density.
Do particles travel along with a sound wave?	No. They vibrate about mean positions; only the disturbance and energy travel.

Likely question	Model answer
State the relation between v, λ and ν.	$v = \lambda \times \nu$; and $\nu = 1/T$.
Why does sound travel faster in solids than in gases?	Particles in solids are closer and pass the disturbance quicker; speeds are about 5000 m s^{-1} in steel against 340 m s^{-1} in air.
Define pitch and loudness. What do they depend on?	Pitch is how frequency is perceived (high frequency = shrill); loudness is how amplitude is perceived.
What is the audible range of humans?	20 Hz to 20 kHz.
Why is the minimum distance for hearing an echo 17 m?	Sounds must be 0.1 s apart; in 0.1 s sound covers 34 m to and fro, so the reflector is 34/2 = 17 m away.
Differentiate between echo and reverberation.	Echo is a separate repeated sound after at least 0.1 s; reverberation is a prolonged sound from reflections less than 0.05 s apart.
How does a bat find its prey in the dark?	By echolocation: it emits ultrasound and senses the echoes from obstacles and prey.
In a film a spaceship explodes with a flash and a bang together. What is wrong?	Sound cannot travel in the vacuum of space, and light would be seen before any sound anyway.

★ KEY TAKEAWAYS

- Sound is produced by **vibrations** and is a **longitudinal mechanical wave** needing a medium; it cannot travel in **vacuum**.
- The **disturbance travels, not the particles**; a wave is a series of **compressions and rarefactions** carrying energy.
- **$v = \lambda\nu$** and **$\nu = 1/T$**; speed depends on the medium (**solids > liquids > gases**) and rises with temperature in air.
- **Pitch ↔ frequency, loudness ↔ amplitude, timbre ↔ overtones**; audible range **20 Hz–20 kHz**.
- **Echo**: reflector at least **17 m** away (0.1 s); **reverberation**: gaps under 0.05 s.
- **Ultrasound** is used in sonar, echolocation, imaging and cleaning; **infrasound** helps detect earthquakes and storms.

CHAPTER 09 · BIOLOGY

Cell: The Building Block of Life

All living things are made of cells. This chapter shows how cells are studied with microscopes, what a cell is made of (membrane, wall, cytoplasm, nucleus and organelles), how water moves in and out by osmosis, how cells divide by mitosis and meiosis, and how the Cell Theory unites all of biology.

Big Questions

- Where does a cell come from?
- How is the cell the structural and functional unit of life?
- How does a cell multiply?

9.1 Origin of Life and Levels of Organisation

- Scientists widely accept that life began in water, possibly in small pools with changing conditions, such as **hot springs**. The hot springs of **Puga Valley (Ladakh)** stay near boiling point even in cold air, resembling early Earth (about 3.5 billion years ago). They hold heat-loving, single-celled bacteria called **thermophiles**.
- Researchers of the Birbal Sahni Institute of Palaeosciences (Lucknow) found **calcium carbonate** deposits forming rapidly around them. Such deposits may have shielded early organic molecules and helped form the first protective membrane.
- **Unicellular** organisms (bacteria, yeast) are one cell; **multicellular** ones (plants, fish, birds, humans) have millions of cells working together.

Level	Meaning	Example
Cell	Basic structural and functional unit	A nerve cell
Tissue	Group of similar cells doing similar work	Muscle tissue
Organ	Different tissues organised for a function	Lungs
Organ system	Several organs working together	Respiratory system: nasal pores, nasal cavity, trachea, lungs

9.2 How to Study Cells?

- The human eye can separate two points only if they are about **0.1 mm** apart (seen from the near point, 25 cm). This is its **limit of resolution**. Most cells are smaller, so tools are needed.
- **Robert Hooke (1665)** first saw cells, in a thin slice of **cork**, with a self-made microscope of 200–300X. The box-like compartments he named 'cells'.
- A **light microscope** uses convex lenses: an **objective** lens (e.g. 10X, 40X) and an **eyepiece**. Parts: eyepiece, body tube, objective lens, coarse and fine adjustment knobs, stage, mirror, handle, base.
- **Electron microscopes** use a beam of electrons instead of light and reveal details at the nanometre scale (1 nm = one-billionth of a metre); a scanning electron microscope image can show structures like the **stomata** of a Colocasia leaf.

- Microscopes have improved in **resolution** (clarity), **contrast** (brightness difference between parts) and **magnification**.

FORMULA

Total magnification = eyepiece power × objective power (10X × 10X = 100X). Size of a cell = diameter of field of view (µm) ÷ number of cells along the diameter. 1 mm = 1000 µm.

Activity 9.1 — estimate cell size
Place a mm ruler on the stage and measure the diameter of the field of view (say 5 mm = 5000 µm).

Count cells
Replace it with an onion-peel slide and count the cells across the diameter in a straight line (say 25).

Calculate
Cell size = 5000 µm ÷ 25 = 200 µm. At 100X it appears 100 times larger.

9.3 Cell Membrane and Osmosis

The **cell membrane (plasma membrane)** is the thin outer boundary of every cell. It protects the contents, gives the cell its individuality and is **selectively permeable** — it lets some substances in or out and blocks others (as in the alveoli of lungs, where O_2 and CO_2 cross).

Activity — potato in water and solution
Cut a potato into two equal pieces, weigh each, and place one in plain water and the other in 20% salt or sugar solution for about an hour.

Observe
The piece in water **swells** (weight rises); the piece in solution **shrinks** (weight falls).

Infer
Water moves through the membrane, but sugar and salt do not. Equal size and initial weights are needed for a fair comparison.

DEFINITION

Diffusion: net movement of particles from higher to lower concentration (no membrane needed).
Osmosis: diffusion of **water** across a selectively permeable membrane, from a dilute solution (more water) to a concentrated one (less water) until the concentrations equalise. Plant roots absorb soil water by osmosis.

Outside solution	Condition	Effect on cell
Isotonic	Solute outside = solute inside	No net water movement; size unchanged
Hypotonic	Solute outside < solute inside (dilute)	Water enters; cell swells
Hypertonic	Solute outside > solute inside (concentrated)	Water leaves; cell shrinks

▸ Structure of the membrane — fluid-mosaic model

- Very thin: about **7–10 nm**; made of **lipids and proteins**.
- A **lipid bilayer** (two layers of fat molecules with water-loving heads outward and water-repelling tails inward) with **proteins embedded** in it.
- Molecules can move sideways, flip and rotate, so it is **fluid**; they are arranged like tiles of a **mosaic**; proteins act as **gatekeepers** for passage.

9.4 Cell Wall

- Plants, fungi and bacteria have an extra layer, the **cell wall**, outside the membrane. Animal cells lack it.
- Plants cannot move, so the wall lets them withstand wind and rain, keeps leaves and flowers firm and holds the plant upright.
- It is rigid but **freely permeable** to water and dissolved minerals, helping roots absorb nutrients. Plant walls are mainly **cellulose** (many glucose units); cellulose in our diet is **roughage**.

Observation	Plant cell (onion / Rhoeo)	Animal cell (cheek)
Shape	Box-like, regularly arranged	Irregular, flexible
In 20% sugar solution	Outline unchanged; inner content shrinks as the membrane pulls away from the wall	Whole cell shrinks
Reason	Rigid cell wall holds shape	No cell wall

Slide method: peel of onion or Rhoeo stained with **safranin**; cheek cells scraped with a toothpick and stained with **methylene blue**. Flexibility of animal cells supports movement of their tissues.

9.5 The Cell Interior — A Coordinated System

Most cells have a membrane, a semi-fluid **cytoplasm** and a **nucleus**. The cytoplasm holds **organelles** (mostly visible only with electron microscopes) and **inclusions** such as starch in plants or crystals of calcium oxalate or silica. In eukaryotes a network of fibres, the **cytoskeleton**, gives support, shape, movement and internal transport.

Feature	Prokaryotic cell	Eukaryotic cell
Meaning	Pro = primitive, karyon = nucleus	Eu = true, karyon = nucleus
Nucleus	Absent; DNA is a single circular molecule in the **nucleoid** (no membrane)	Well-defined, membrane-bound
Membrane-bound organelles	Absent; activities happen in cytoplasm	Present
Typical diameter	1–10 µm	10–100 µm
Number of cells	Usually unicellular	Unicellular or multicellular
Example	Bacteria	Plant and animal cells

> **REMEMBER**
> Viruses, viroids and prions are **acellular** infectious agents, too small for a light microscope. A virus is genetic material in a protein coat; a viroid has no coat; a prion is a misfolded protein with no genetic material.

▸ Why organelles? — the cell as a tiny factory

Separate organelles carry out different jobs at the same time: building materials, removing waste and providing energy.

Organelle	Structure	Function
Nucleus	Double-layered nuclear membrane with pores; nucleolus (makes ribosome subunits); chromatin / chromosomes	Controls the cell; holds DNA (genes) for inheritance
Ribosomes	Tiny; free in cytoplasm or attached to ER	Protein synthesis ('protein factories')
Rough ER (RER)	Network continuous with nuclear envelope; ribosomes on surface	Protein synthesis and secretion (e.g. pancreatic gland cells)
Smooth ER (SER)	No ribosomes	Makes and stores fats (lipids) and hormones
Golgi apparatus	Stacks of flattened sacs; found in 1898 by Camillo Golgi in barn-owl nerve cells	Modifies, sorts and packs proteins/lipids into vesicles ('post office'); forms lysosomes
Lysosomes	Single-membrane sacs full of enzymes	Digest wastes, worn-out parts and foreign matter ('clean-up'); sperm lysosomal enzymes help penetrate the egg
Mitochondria	Double membrane; outer smooth, inner folded into **cristae**; own DNA and ribosomes	Cellular respiration; release energy stored as **ATP** ('powerhouse')
Plastids	Double membrane in plants (chloroplast has stroma and chlorophyll discs)	Make and store food
Vacuole	Large central one in plants with **cell sap**; small in animals	Stores water, minerals, sugars and wastes; keeps cell firm (turgor)

- **Chromatin** is the entangled thread-like DNA material in a non-dividing cell; it condenses into rod-like **chromosomes** (DNA + proteins) just before division. A **gene** is a functional segment of DNA.
- Mature **RBCs** have no nucleus (enucleate): more room for haemoglobin and so more oxygen carried, but they cannot divide or repair themselves and live only about **120 days**.
- Mitochondria and chloroplasts have their **own DNA and ribosomes**, like bacteria, hinting at a shared evolutionary history.
- **Wilting**: when a plant lacks water the vacuole loses water, cells lose firmness and the plant droops.

Plastid	Pigment	Role
Chloroplast	Green chlorophyll	Absorbs sunlight for photosynthesis; sugars and starch stored in stroma
Chromoplast	Yellow, orange or red	Colours flowers and fruits to attract pollinators and seed dispersers
Leucoplast	Colourless	Stores starch, oils or proteins (e.g. in potato and taro)

Structure	Plant cell	Animal cell	Bacterial cell
Cell membrane	Yes	Yes	Yes
Cell wall	Yes	No	Yes
Cytoplasm	Yes	Yes	Yes
True nucleus	Yes	Yes	No (nucleoid)
Membrane-bound organelles	Yes	Yes	No

Structure	Plant cell	Animal cell	Bacterial cell
Plastids / large vacuole	Yes	No (small vacuoles)	No

IMPORTANT

Venter's 2010 experiment: DNA of *Mycoplasma mycoides* was synthesised in the lab and inserted into a related bacterium whose own DNA was removed. The cell then grew and divided as per the new DNA, showing that **DNA controls the cell** — but only the DNA was synthetic, not the whole cell.

9.6 Cell Growth and Division

The body grows not because cells get bigger without limit but because cells **divide**. Cell division forms new cells from pre-existing ones for **growth, repair and reproduction**. About 1% of body cells (hundreds of billions) are replaced daily. The **onion root tip** is a good place to see dividing cells (stages differ in appearance).

Prepare roots

Grow onion roots in water for 5–6 days; cut 2–3 cm tips and fix in aceto-alcohol (acetic acid : ethanol = 1 : 3) for 24 hours; store in 70% ethanol.

Soften and stain

Place a root on a slide, add dilute HCl for 10–15 min, rinse, then stain with aceto-carmine for 5–10 min and warm gently.

Squash and observe

Cut the tip, add a coverslip, press gently with the thumb and view: cells show different stages of division.

Feature	Mitosis	Meiosis
Occurs in	Body (somatic) cells	Only reproductive organs: testes/ovaries; anthers/ovary of flowers
Divisions	One	Two, one after the other
Daughter cells	Two	Four
Chromosome number	Same as parent	**Halved** in each gamete
Genetic nature	Identical to parent	Varied
Purpose	Growth, repair, maintenance, asexual reproduction	Gametes for sexual reproduction; diversity

- Every human starts as one fertilised egg that divides by mitosis into trillions of cells. Prokaryotes also divide; eukaryotes follow a controlled **cell cycle**.
- In **fertilisation** two gametes fuse and restore the original chromosome number; this variation is why children resemble but differ from parents.
- **Errors**: faulty mitosis → uncontrolled division, tumours, abnormal chromosome numbers; faulty meiosis → genetic disorders, developmental problems, early pregnancy loss, reduced fertility.
- **Cell culture**: cells grown outside the body in nutrient medium under right temperature, pH, moisture and sterility, used for studying cells and making biochemicals, food, medicines and vaccines.

9.7 Cell Theory

Scientist	Year	Contribution
Matthias Schleiden	1838	All plants are made of cells
Theodor Schwann	1839	All animals are made of cells
Rudolf Virchow	1855	New cells arise only from pre-existing cells

DEFINITION

Cell Theory: (1) all living organisms are made of one or more cells; (2) the cell is the basic unit of structure and function; (3) all cells arise from pre-existing cells.

- Haberlandt (1902) proposed **totipotency**: a living plant cell can grow into a complete plant, founding plant tissue culture.
- Every cell has a definite **life span**. Many animal cells stop dividing on touching neighbours (**contact inhibition**). **Cancer cells** lose this control and form **tumours** (benign or malignant; malignant ones invade and spread). Plant cells show no contact inhibition because of their rigid walls.
- **Programmed cell death (PCD)** is genetically regulated destruction of cells, needed for development, quality control and immunity (e.g. it removes cells between embryonic fingers; without it hands would be webbed).

9.8 Quick Exam Questions

Likely question	Model answer
Who first observed cells and how?	Robert Hooke in 1665, in a slice of cork under a self-made microscope; he named the box-like compartments cells.
What is osmosis? Why does a potato piece shrink in strong salt solution?	Osmosis is movement of water across a selectively permeable membrane from dilute to concentrated solution; the strong solution is hypertonic, so water leaves the cells.
Differentiate isotonic, hypotonic and hypertonic solutions.	Isotonic: equal solute outside and inside, no change; hypotonic: less solute outside, cell swells; hypertonic: more solute outside, cell shrinks.
Why does a plant cell not burst or shrink in size in a solution?	The rigid permeable cell wall holds its shape; only the inner content changes.
Explain the fluid-mosaic model.	A lipid bilayer with embedded proteins; molecules move sideways (fluid) and lie like mosaic tiles; proteins help substances pass.
Compare prokaryotic and eukaryotic cells.	Prokaryotes: nucleoid, no membrane-bound organelles, 1–10 µm, usually unicellular; eukaryotes: true nucleus and organelles, 10–100 µm.
Why are mitochondria called the powerhouse of the cell?	They carry out cellular respiration and store the energy as ATP, the cell's energy currency.
Differentiate RER and SER.	RER has ribosomes and makes proteins; SER has none and makes fats and hormones.
Distinguish chloroplast, chromoplast and leucoplast.	Green photosynthetic pigment; yellow/orange/red pigments for colour; colourless, stores food.

Likely question	Model answer
Why do RBCs lack a nucleus and what is the effect?	More space for haemoglobin and oxygen; but they cannot repair or divide, living about 120 days.
Compare mitosis and meiosis.	Mitosis gives two identical cells with the same chromosome number for growth and repair; meiosis gives four cells with half the number, forming gametes and variation.
State the Cell Theory.	All organisms consist of cells; the cell is the basic unit; all cells come from pre-existing cells.
What is the link between wilting and vacuoles?	Without water the vacuole shrinks, pressure falls and cells lose firmness, so the plant droops.

★ KEY TAKEAWAYS

- The **cell** is the basic structural and functional unit of life; Hooke saw cells in **cork (1665)**.
- The membrane is **selectively permeable** (fluid-mosaic); **osmosis** is diffusion of water, giving **isotonic, hypotonic, hypertonic** effects.
- **Cell wall** (cellulose) occurs in plants, fungi and bacteria; **prokaryotes** have a nucleoid, **eukaryotes** a true nucleus and organelles.
- Learn organelle functions: nucleus (control), ribosome (proteins), ER, Golgi (packing), lysosome (digestion), **mitochondria (ATP)**, plastids (food), vacuole (storage).
- **Mitosis**: 2 identical cells; **meiosis**: 4 cells with half the chromosomes.
- **Cell Theory** (Schleiden, Schwann, Virchow); cancer is uncontrolled division, while **PCD** removes cells in an orderly way.

CHAPTER 10 · BIOLOGY

Tissues in Action

Cells of the same kind group into tissues, giving division of labour and efficiency. This chapter classifies plant tissues (meristematic and permanent) and animal tissues (epithelial, connective, muscular, nervous), then links bones, muscles, joints and the skeleton to movement.

Big Questions

- How are tissues in plants and animals different, and why?
- Why do some tissues grow throughout life while others do not?
- How is the structure of a tissue matched to its function?

10.1 Tissues and Why Plants and Animals Differ

- A **tissue** is a group of cells, similar in structure, that work together for a specific function. Organisation: cells → tissues → organs → organ systems → organism. Unicellular organisms like *Amoeba* do everything in one cell; multicellular ones show **division of labour**, which raises efficiency.
- Animal examples: muscle tissue moves the body, nervous tissue carries messages. Plant examples: **xylem** carries water and minerals, **phloem** carries food.
- **Why plant and animal tissues differ**: plants are fixed in place and need rigidity (cell walls and supporting tissues); animals move and have flexible cells. Plants make food by photosynthesis, animals digest it. Transport and growth patterns also differ (plant growth is localised in meristems).

10.2 Tissues for Growth in Plants — Meristematic Tissue

Plants grow in length, in girth and by regrowth after cutting or grazing. This needs actively dividing cells, forming **meristematic tissue**.

Activity 10.1 — onion roots
Place two onion bulbs over water-filled jars and measure root length on days 1–3.

Cut
On day 3 cut about 1 cm off the root tips in Jar B and keep measuring.

Result
Roots in Jar A keep growing; Jar B roots stop. Hence roots grow from their tips, where cells divide continuously (growth zone).

Meristem	Location	Gives growth in	Examples / notes
Apical	Tips of roots and shoots	Length (height, root depth)	Root tip of onion; sapling
Lateral	Ring of dividing cells in the stem circumference	Girth (thickness); makes new cells inside and outside concentrically	Dicot stems; **annual rings** in a trunk
Intercalary	Base of internode / just above the node	Regrowth and branching after cutting	Grass after mowing or grazing; hedges becoming bushy

- **Annual rings**: wide ring = favourable year, narrow = unfavourable. Counting them gives the tree's age and past climate (a teak of 40 years shows 40 rings).

- **Node** is where leaves or branches arise; **internode** lies between two nodes. If a young stem tip is cut, length growth stops but new branches arise at nodes.
- **Features of meristematic cells**: small, thin walls, large prominent nucleus, dense cytoplasm with many organelles, **no vacuoles**, tightly packed with no intercellular spaces — suited to fast, continuous division.
- **Differentiation**: some new cells lose the power to divide and become specialised; this converts meristematic into **permanent tissue**.

10.3 Permanent Tissues

Permanent tissues may be **simple** (one type of cell) or **complex** (more than one type of cell). Functionally they protect, support or conduct. A sunflower stem T.S. shows epidermis, ground tissue (parenchyma, collenchyma, sclerenchyma) and vascular tissue (xylem and phloem).

▸ Protective tissue — epidermis

- Outermost layer: a single layer of tightly packed, flat rectangular cells, coated with waxy **cutin** forming the **cuticle**. It guards against injury, water loss, parasites and extreme conditions; desert plants have a thicker cuticle to cut transpiration.
- **Root hairs** (epidermal projections) increase the surface for absorbing water and minerals.
- In leaves the epidermis has **stomata** for gas exchange and **transpiration** (loss of water vapour), which creates a transpiration pull in xylem and removes wastes.
- In older stems, cells below the epidermis form **cork cambium** (a lateral meristem) that makes **cork cells** — dead, compact and impermeable to water and gases — which form the **bark**.

▸ Supporting tissue — simple permanent tissues

Tissue	Cells and wall	Living?	Function / where
Parenchyma	Thin walls, loosely packed with intercellular spaces	Living	Stores food; photosynthesis in green parts (chlorenchyma); air spaces help aquatic plants float
Collenchyma	Corners unevenly thickened with **pectin**	Living	Support with **flexibility**; stems, leaf stalks, tendrils bend without breaking
Sclerenchyma	Thick walls hardened with **lignin**	Mostly dead	Hardness and strength; woody parts, leaf veins, coconut husk fibres, seed coats, walnut shell

▸ Conducting tissue — complex permanent tissues (vascular)

Feature	Xylem	Phloem
Carries	Water and minerals from roots upward	Food (sugars) from leaves to all parts
Components	Tracheids, vessels, xylem parenchyma, xylem fibres	Sieve tubes, companion cells, phloem parenchyma, phloem fibres
Living / dead	Mostly dead (only xylem parenchyma is living); tubular, thick-walled	Mostly living
Extra role	Gives mechanical strength	Companion cells regulate sieve tubes and load/unload sugars; parenchyma stores food, resin, tannins and latex

Sieve tubes are long cells joined end to end by perforated walls (sieve plates). Fibres in both tissues are sclerenchymatous and give strength.

Tissue system	Made of	Role
Dermal	Epidermis (and cork)	Outer covering, protection, reduces water loss
Ground	Parenchyma, collenchyma, sclerenchyma	Main body between dermal and vascular tissue
Vascular	Xylem and phloem	Conduction

IMPORTANT

Meristematic tissues divide; permanent tissues have lost this ability. Plant tissue types: **meristematic** (apical, lateral, intercalary) and **permanent** (simple: parenchyma, collenchyma, sclerenchyma; complex: xylem, phloem; protective: epidermis, cork).

10.4 Animal Tissues

Animal tissues are of four main kinds: **epithelial, connective, muscular and nervous**.

▸ Epithelial tissue

Covers the body (skin) and lines internal organs (mouth, lungs, blood vessels, intestine). Cells are packed tightly with little space, blocking germs, reducing water loss and aiding absorption, secretion and movement of substances.

Function	Structure	Location
Exchange (rapid diffusion of gases/liquids)	Single layer of thin, flat cells	Blood vessels, lungs (alveoli)
Protection from injury, friction, microbes	Many layers; outer cells flat and tight	Skin, mouth, oesophagus
Secretion (mucus, enzymes, hormones, sweat, saliva)	Cuboidal or columnar gland cells	Salivary glands, sweat glands, stomach lining
Sensing (smell, taste, sound, balance)	Receptor cells with hair-like cilia	Nostrils, taste buds, inner ear
Absorption of nutrients and water	Single layer of tall, pillar-like cells, often with hair-like projections	Small intestine

▸ Connective tissue

Connects and supports other tissues. The **matrix** (the non-cellular material) decides its nature: watery and jelly-like in blood, hard and rigid in bone.

Tissue	Matrix / nature	Function
Blood	Fluid plasma (about 55% of volume) with formed elements — RBC, WBC, platelets (45%)	Transports nutrients, gases, hormones
Bone	Hard, rigid matrix with calcium and phosphorus compounds	Strength, support, protection
Cartilage	Soft, jelly-like, flexible (ear, nose tip, rib ends, between vertebrae)	Flexibility; cushions bone ends against shock
Tendon	Tough flexible band	Joins **muscle to bone**; transmits force

Tissue	Matrix / nature	Function
Ligament	Tough, slightly elastic band	Joins **bone to bone**; stabilises joint, limits movement, prevents dislocation

Blood component	Role
RBC (haemoglobin, an iron-rich protein)	Red colour; carries oxygen; lives about 4 months
Platelets	Blood clotting at a cut
WBC	Gather at infection, causing pus, redness and swelling; defence

▸ Muscular tissue

Muscle	Control	Cell features	Location / action
Skeletal	Voluntary	Long cylindrical, unbranched, **multinucleate**, **striated** fibres	Attached to skeleton; running, writing, lifting
Smooth	Involuntary	Spindle-shaped, single nucleus, **no striations**	Stomach, intestine; slow continuous movement (digestion)
Cardiac	Involuntary	Cylindrical, **branched**, one nucleus, faint striations	Heart only; beats rhythmically throughout life without fatigue

▸ Nervous tissue

Made of **neurons**, specialised to receive, process and transmit messages; the brain is the control centre. Muscles need instructions from nervous tissue (e.g. the brain makes the heart beat faster in exercise).

Part of neuron	Function
Cell body	Contains the nucleus; controls the cell
Dendrites	Receive signals from other neurons
Axon	Long fibre carrying the message away; ends in axon terminals that pass it on to other cells

10.5 Musculoskeletal System and Joints

- Made of bones, muscles, joints, cartilage, tendons and ligaments; helps us stand, move, hold posture and protect organs; it works under the **nervous system**. Muscles pull on bones via tendons, producing movement at a joint.
- The adult skeleton is about **12–15%** of body weight; muscle about 40–50% in men and 30–40% in women (approximate).

DEFINITION

Joint: the junction between two or more bones. Joints permit movement, but cannot move bones themselves — muscles do.

Joint	Movement allowed	Example
Ball and socket	Free: forward, backward, sideways, circular	Shoulder (round head of upper arm bone fits in hollow of shoulder bone); hip

Joint	Movement allowed	Example
Hinge	One direction only, like a door hinge	Elbow; knee (kneecap protects)
Pivot	Side-to-side rotation	Skull on backbone (head turning 'no')
Fixed	No movement	Bones of the skull, protecting brain, eyes and ears

10.6 Skeletal System

- Framework of bones for strength and protection: **skull, vertebral column and rib cage**.
- The **backbone (spine)** is a flexible column of **vertebrae**, with cartilage discs between them that cushion, give flexibility for bending and twisting and protect the spinal cord.
- **Rib cage**: 12 pairs of ribs joined at the back to the spine and in front to the **sternum** by flexible cartilage. It protects heart and lungs and expands and contracts during breathing.
- Good posture, nutrition, exercise and **yoga** (International Yoga Day, 21 June) keep bones strong, muscles fit and joints flexible.

10.7 Totipotency and Other Extensions

- **F. C. Steward (1958)** grew single phloem cells of carrot in a nutrient medium with sugars and hormones; they divided into a cell mass (dedifferentiation) and then redifferentiated into roots, shoots and a whole plant. Such cells are **totipotent**, like a zygote.
- Cork cambium and cork also arise in old stems (see protective tissue).
- **Stem cells** in bone marrow divide to make new cells; bone marrow transplants help patients with leukemia and thalassemia.
- Crown gall disease (tumour-like swelling) is caused by *Agrobacterium tumefaciens*; its gene transfer is now used in genetic engineering of plants.

10.8 Quick Exam Questions

Likely question	Model answer
What is a tissue? Why is division of labour useful?	A group of similar cells working together for a function; specialisation makes complex life processes efficient.
Why do plants have more supporting tissue than animals?	They are fixed in one place and must stand upright against wind and rain; animals move and have flexible cells.
Name the three types of meristem with location and function.	Apical: tips, length; lateral: along the stem circumference, girth; intercalary: base of internodes/nodes, regrowth.
Why do meristematic cells lack vacuoles?	They need dense cytoplasm and a large nucleus for fast division; vacuoles would crowd them.
How do plant growth rings form and what do they tell?	Lateral meristem adds new cells yearly; the number of rings gives age and the width shows climate of each year.
Differentiate parenchyma, collenchyma, sclerenchyma.	Parenchyma: thin-walled living storage cells; collenchyma: pectin-thickened corners, flexible support; sclerenchyma: lignified, dead, hard.

Likely question	Model answer
Why can a fresh twig bend but a dry twig snaps?	Collenchyma keeps the fresh twig flexible; dry twigs rely on rigid lignified tissue.
Compare xylem and phloem.	Xylem carries water and minerals upwards and is mostly dead; phloem carries food and is mostly living with sieve tubes and companion cells.
What is the function of stomata and cuticle?	Stomata allow gas exchange and transpiration; cuticle reduces water loss and protects.
Why are RBCs red and what do platelets do?	Haemoglobin (iron-rich) makes them red; platelets help clotting.
Differentiate tendon and ligament.	Tendon joins muscle to bone; ligament joins bone to bone.
Compare skeletal, smooth and cardiac muscle.	Skeletal: voluntary, striated, multinucleate; smooth: involuntary, spindle, no striations; cardiac: involuntary, branched, heart only.
Name the parts of a neuron.	Cell body, dendrites and axon with axon terminals.
Name four types of joints with an example each.	Ball and socket (shoulder), hinge (elbow/knee), pivot (neck-skull), fixed (skull).
What is totipotency?	Ability of a single cell to grow into a complete organism, as shown with carrot phloem cells by Steward.

★ KEY TAKEAWAYS

- Tissues are groups of similar cells; **plants** have **meristematic** (apical, lateral, intercalary) and **permanent** tissues.
- Permanent tissues: **epidermis** (protection), **parenchyma, collenchyma, sclerenchyma** (support), **xylem and phloem** (conduction).
- Animal tissues: **epithelial, connective (blood, bone, cartilage, tendon, ligament), muscular (skeletal, smooth, cardiac), nervous (neuron)**.
- **Tendon**: muscle–bone; **ligament**: bone–bone.
- Joints: **ball and socket, hinge, pivot, fixed**; the skeleton comprises skull, backbone and rib cage.
- **Totipotent** plant cells (Steward, carrot) can regenerate a whole plant.

CHAPTER 11 · BIOLOGY

Patterns in Life: Diversity and Classification

Earth holds an immense variety of life. To study it, scientists group organisms by shared features and evolutionary links. This chapter covers biodiversity, how classification evolved, the Five Kingdom system, the groups of plants and animals, the hierarchy of ranks, binomial naming, fossils and threats to biodiversity.

Big Questions

- What is biodiversity and why does it matter?
- On what basis are plants and animals classified?
- How does classification help in farming and conservation?

11.1 Biodiversity and India as a Hotspot

- **Biodiversity** is the variety of life forms in all habitats. It keeps nature stable: ocean algae release much of our oxygen, fungi and bacteria decompose waste and make soil fertile, birds, bees and bats pollinate, plants capture sunlight to feed nearly all life. Humans depend on it for food, shelter, medicine and livelihood.
- Farmers conserved crop varieties with traits such as drought tolerance, pest resistance and growth in poor soil; **diversity reduces the risk of crop failure** and strengthens food security.
- **Endemic species** are found naturally only in one region: Nilgiri tahr, Lion-tailed macaque, *Nepenthes khasiana* (Indian pitcher plant), Neelakurinji.
- **Biodiversity hotspot**: a region with many endemic species that has suffered major habitat loss. Hotspots touching India: **Western Ghats, Indo-Burma (incl. North-East India), Himalayas, Sundaland (incl. Nicobar Islands)**.
- India's varied landscapes (mountains, desert, rainforest, plateau, coasts) support many species. Ancient Indian ideas such as the Sangam *Tinai* landscape classification and sacred groves protected local habitats.

Present biodiversity is the outcome of small differences among individuals that helped some survive and reproduce, accumulated over many generations.

11.2 Why and How to Classify

The same organism can fit different groups depending on the **criterion** used (for example by eating habit: carnivore, herbivore; or by activity: diurnal, nocturnal). Scientists therefore choose features that reflect real relationships. Using the library analogy, books must be arranged to be found.

Criterion	What is looked at
External features	Shape, size, body organisation
Mode of nutrition	Autotrophic or heterotrophic
Internal structure	Skeleton, organs, tissues
Cell structure	Unicellular/multicellular, prokaryote/eukaryote, cell wall present or absent

Criterion	What is looked at
Ecological role	Producer, consumer, decomposer
Reproduction	Asexual and/or sexual
Genetic similarity	DNA similarities

IMPORTANT

Similar features suggest **common ancestry**. **Biological classification** is the scientific system of grouping organisms by similarities and differences.

▸ Uses of classification

- Makes study organised and systematic.
- Shows similarities, differences and relationships among organisms.
- Helps identify and name new organisms; gives scientists worldwide a common language.
- Supports conservation by identifying threatened organisms.
- Case study — **Pakke Tiger Reserve** (Arunachal Pradesh): about 300 bird species (India has about 1,300) and four hornbills (Rufous-necked, Oriental Pied, Great, Wreathed) which need large old cavity trees and specific fruits, so classifying helps study distribution and links; losing old trees would endanger them.

11.3 Classification Systems Over Time

Scientist / year	System	Groups
Aristotle (4th c. BCE)	Artificial; by habitat/appearance	Land, water, air animals (limited: only visible features). Rigveda and Brihat Samhita also grouped animals by habitat
Linnaeus (18th c.)	Two kingdom	Plantae (fixed, make own food) and Animalia (move, depend on others). Problem: *Amoeba*, *Paramecium*, bacteria
Haeckel (1866)	Three kingdom	Added **Protista** for unicellular microscopic life
Copeland (1938)	Four kingdom	Monera, Protista, Plantae, Animalia (bacteria lack a true nucleus)
Whittaker (1969)	**Five kingdom**	Monera, Protista, **Fungi**, Plantae, Animalia (fungi are heterotrophic absorbers, not plants)
Carl Woese (1977)	Three domain (genetic data)	Bacteria, Archaea, Eukarya

Classification keeps changing as tools (microscopes, staining, DNA studies) improve: science is an ongoing process of reasoning.

11.4 The Five Kingdom Classification

Basis: **cell type** (prokaryote or eukaryote), **cell structure** (cell wall present or not), **level of organisation** (unicellular or multicellular) and **mode of nutrition** (autotrophic or heterotrophic).

Kingdom	Cell type	Body	Cell wall	Nutrition	Examples
Monera	Prokaryote	Unicellular	Present	Auto- or heterotrophic	Bacteria, archaea, cyanobacteria
Protista	Eukaryote	Unicellular	Absent or cellulose	Auto- or heterotrophic	*Amoeba*, *Paramecium*, *Euglena*, *Chlamydomonas*
Fungi	Eukaryote	Mostly multicellular (yeast unicellular)	**Chitin**	Heterotrophic (absorb)	Mushroom, yeast, *Aspergillus*, *Penicillium*, bread mould
Plantae	Eukaryote	Multicellular	**Cellulose**	Autotrophic (photosynthesis)	Mosses, ferns, pines, roses
Animalia	Eukaryote	Multicellular	Absent	Heterotrophic	Ant, frog, bird, human

▸ Monera

- Found everywhere: soil, water, air, hot springs, inside bodies, in the gut of ruminants (source of biogas from dung).
- Harmful ones are pathogens; useful ones include *Lactobacillus* and *Rhizobium*; some break down oil, pesticides and sewage.
- **Cyanobacteria** (blue-green algae) are autotrophs and were among the first to release oxygen (~2.5 billion years ago); fossils form **stromatolites** (found in Rajasthan and Madhya Pradesh).
- Ram Bux Singh, 'Father of Modern Biogas', set up India's first scientific biogas plant in 1957 at Ramnagar, Sitapur.

▸ Protista

Microscopic single-celled eukaryotes of water and moist places (found in a **hay infusion**: amoeba, paramecium, euglena). Some are autotrophs, others heterotrophs; they are links in aquatic food chains, produce oxygen, feed small animals or act as decomposers.

▸ Fungi

- Cell wall of **chitin**; absorb food from dead and decaying matter through fine filaments forming a **mycelium**; mostly **saprophytes** (decomposers recycling nutrients, keeping soil fertile). Some are mutualistic partners, some are parasites.
- Reproduce sexually and asexually, often by **spores**; best in warm, moist conditions.
- Uses: *Aspergillus* and *Penicillium* give enzymes and antibiotics; mushroom farming needs little space and takes 30–45 days. Folk taxonomy helps tribal communities tell edible from poisonous mushrooms.

▸ Plantae — five classes

Class	Body	Water needed?	Seeds / flowers	Examples
Thallophyta (algae)	Undifferentiated **thallus**; mostly aquatic	Live in water	None	*Spirogyra*
Bryophyta	Slightly differentiated; **rhizoids**, no vascular tissue	Needs water for reproduction; 'amphibians of the plant kingdom'	None	Moss, *Marchantia*
Pteridophyta	True roots, stem, leaves; **xylem and phloem**	Needs water for reproduction	No seeds	Ferns

Class	Body	Water needed?	Seeds / flowers	Examples
Gymnosperm	Needle/scale leaves reduce water loss; cold, dry regions	Not needed for fertilisation	**Naked seeds** on cones, no fruit	Pine, cycads
Angiosperm	Most complex; well-developed roots, stem, leaves	Not needed	**Flowers, fruits** enclosing seeds; most diverse	Gulmohar, rose, mango

- Flowers attract pollinators; fruits aid seed dispersal (animals, wind, water). Angiosperms can be grouped into monocots and dicots by leaf venation and other features.
- **Lichens** (e.g. *patthar ke phool*) are a symbiosis of an alga (makes food) and a fungus (protects); they are **bioindicators of air quality** and are used for dyes, spice and medicine; some are poisonous.
- Trend: plants evolved from water dependence to transport tissue, seeds and flowers, adapting to land.
- *Hortus Malabaricus* (17th c., Hendrik van Rheede with Itty Achudan) recorded hundreds of Indian medicinal plants.

▸ Animalia

Multicellular, heterotrophic, with locomotion and quick response to stimuli. The main criterion is the **notochord** (a flexible rod-shaped structure that can become the vertebral column): animals are **non-chordata (invertebrates)** or **chordata** (protochordates and vertebrates).

Phylum	Key features	Examples
Porifera (pore-bearers)	Multicellular, **no tissues/organs**; pores let water carry food and oxygen; fixed, aquatic	Sponges (1 kg can filter up to 24,000 L water a day)
Cnidaria	True **tissues**; tentacles capture prey; one opening for food and waste	*Hydra*, jellyfish, corals
Platyhelminthes	**Bilateral symmetry**, flat body, one opening; many parasitic with hooks and suckers	Flatworms (tapeworm)
Nematoda	Cylindrical; **two openings** (mouth and anus); organ system level; separate sexes	Roundworms
Annelida	**Segmented** body, body cavity, muscles and nerve cord; organ systems	Earthworm
Arthropoda	Jointed appendages, segmented body, hard **external skeleton** (protection, less water loss)	Insects, crabs, spiders
Mollusca	Soft body, often a protective shell; head, muscular foot, hump	Snail, squid, octopus
Echinodermata	Spiny skin; hard **internal skeleton** of calcium carbonate; no notochord	Starfish, sea urchin

- Trend: simple → tissues → bilateral symmetry → two openings → segmentation → exoskeleton / endoskeleton: more complexity but new challenges.
- **Protochordates** (*Amphioxus*) have a notochord at some life stage — a transition from invertebrates to vertebrates.
- **Vertebrates** have a backbone protecting brain and spinal cord and permitting larger size; five groups: **fish, amphibians, reptiles, birds, mammals**.

- Adaptations: fins and gills (fish), feathers and hollow bones (birds), fat storage (camel), thick fur (polar bear), mammary glands (mammals).

Parasitic worms enter through contaminated water and food, so wash hands, cook food and drink boiled or filtered water. Diverse forests also protect (mangroves reduced damage in the 1999 Odisha super cyclone; diversity lowers Kyasanur Forest Disease risk).

11.5 Hierarchy and Binomial Nomenclature

REMEMBER

Hierarchy from broad to specific: **Kingdom → Phylum → Class → Order → Family → Genus → Species**. Lower down, organisms share more features; each group is part of the one above (like a postal address).

Rank	Tiger	Pea
Kingdom	Animalia	Plantae
Phylum	Chordata (sub-phylum Vertebrata)	Magnoliophyta
Class	Mammalia	Magnoliopsida
Order	Carnivora	Fabales
Family	Felidae	Fabaceae
Genus	*Panthera*	*Pisum*
Species	*P. tigris*	*P. sativum*

Binomial nomenclature (Linnaeus, 18th century) gives each organism a two-part Latin scientific name, avoiding confusion between local names (tiger = bagh, puli, tigre).

- First word = **genus** (capital letter); second = **species** (small letters). *Panthera tigris* (tiger) and *Panthera leo* (lion) share genus *Panthera*; mango = *Mangifera indica*.
- Written in **italics** when printed, **underlined** when handwritten.
- A **species** is a group of similar individuals that can interbreed and produce offspring.

11.6 Fossils and Threats to Biodiversity

- **Fossils** are preserved remains of organisms in layers of rock, sand and mud; older layers hold simpler forms, newer ones more complex forms — evidence of change over time. Birbal Sahni (BSIP, Lucknow) studied fossil plants.
- Threats: **pollution, deforestation, overuse of resources, climate change**. Losing one species can cause dependent species to decline.
- Purple Frog (*Nasikabatrachus sahyadrensis*), found in 2003 in the Western Ghats, an ancient frog family that lives underground and breeds in monsoon.
- Sangai deer of Loktak's floating **phumdis** (Manipur) was thought extinct in 1951 and rediscovered in 1953; habitat loss now threatens it (IUCN Red List).

11.7 Quick Exam Questions

Likely question	Model answer
Define biodiversity, endemic species and biodiversity hotspot.	Biodiversity is variety of life; endemic species occur only in one region; a hotspot has many endemic species and major habitat loss.
Name the biodiversity hotspots of India.	Western Ghats, Indo-Burma (North-East India), Himalayas and Sundaland (Nicobar Islands).
Why do we classify organisms?	For orderly study, to see relationships, name new species, share a common language and plan conservation.
List the criteria used for five kingdom classification.	Cell type, cell wall, level of organisation and mode of nutrition.
Why was the two kingdom system inadequate?	Organisms like *Amoeba* and bacteria are unicellular and fit neither plants nor animals.
Why are fungi placed in a separate kingdom?	They have chitin walls and absorb nutrients (heterotrophic), unlike plants.
Distinguish Monera and Protista.	Monera are prokaryotic with no true nucleus; Protista are single-celled eukaryotes with a nucleus.
Why are bryophytes called amphibians of the plant kingdom?	They live on land but need water for reproduction.
How do pteridophytes differ from bryophytes?	Pteridophytes have true roots, stems, leaves and xylem/phloem; bryophytes have only rhizoids and no vascular tissue.
Differentiate gymnosperms and angiosperms.	Gymnosperms have naked seeds on cones; angiosperms have flowers and seeds enclosed in fruits.
What is a notochord and how does it classify animals?	A flexible rod-like structure; animals lacking it are non-chordates, those with it are chordates.
What is the main feature of Porifera, Cnidaria and Platyhelminthes?	Cellular level (pores), tissue level (tentacles), and bilateral symmetry respectively.
Write the taxonomic hierarchy.	Kingdom, phylum, class, order, family, genus, species.
State the rules of binomial naming with an example.	Genus first (capital), species next (small), italics/underlined: *Mangifera indica*.
Why are viruses not in the five kingdoms?	They are acellular, inactive outside a host and lack cellular organisation, so they fit none of the criteria.

★ KEY TAKEAWAYS

- **Biodiversity** keeps ecosystems stable; India has several **hotspots** with **endemic** species.
- Classification groups organisms by cell type, cell wall, organisation and nutrition; **Whittaker (1969)**: **Monera, Protista, Fungi, Plantae, Animalia**.
- Plants: **Thallophyta, Bryophyta, Pteridophyta, Gymnosperm, Angiosperm** — progression to land, vascular tissue, seeds, flowers.
- Animals: **notochord** divides non-chordates (8 phyla) from chordates; vertebrates have 5 groups.
- Hierarchy: **Kingdom → Phylum → Class → Order → Family → Genus → Species**; **binomial name** = Genus + species (Linnaeus).
- **Fossils** show life's history; pollution, deforestation and climate change threaten biodiversity.

CHAPTER 12 · BIOLOGY

Reproduction: How Life Continues

Reproduction is the biological process by which living beings produce new individuals of their own kind, so life continues even though individuals die. This chapter compares asexual and sexual reproduction, studies vegetative propagation, flowers, pollination and fertilisation, animal strategies, and the human reproductive system, menstrual cycle, pregnancy and reproductive health.

Big Questions

- When does a farmer prefer asexual or sexual methods of crop production?
- Why do complex animals and flowering plants use sexual reproduction while simple organisms reproduce mainly asexually?

12.1 Asexual Reproduction

In **asexual reproduction** a single parent produces offspring that are almost exact copies. In **sexual reproduction** two parents contribute, and offspring inherit a mix of characters, giving variation that helps adaptation and may lead to new species.

Feature	Asexual	Sexual
Parents	One	Two (male and female gametes)
Offspring	Genetically identical (**clones**)	Show variation
Basis	**Mitosis**	**Meiosis** (gametes) + fertilisation
Speed	Fast; quick population rise in favourable conditions	Slower
Examples	Bacteria, amoeba, yeast, hydra, sponge, potato, *Bryophyllum*	Flowering plants, most animals, humans

▸ Vegetative propagation in plants

New plants arise from vegetative parts (growing parts): potato and ginger (underground stems), money plant and sugarcane (stem cuttings), *Bryophyllum* (plantlets on leaf edges).

Method	How it is done	Notes
Cutting	Cut shoot with nodes and internodes (collected in the morning); remove lower leaves; insert half the length in compost-mixed soil at 45–60°; water regularly	Rose, sugarcane, money plant
Grafting	Slit a rooted plant (stock, Plant A); fit a stem piece (scion, Plant B) into it; wrap with cloth/film to keep pests out until it heals	Combines good traits, e.g. fruit trees, roses; KVKs of ICAR train farmers
Layering	Bury the middle of a flexible twig (e.g. lemon) in soil; roots form in 10–15 days; then cut from parent	Plant remains attached to the parent while rooting
Tissue culture	Grow plantlets from shoot tip (apical meristem) in nutrient medium	Mass-produces healthy, virus-free banana plantlets; high yield

▸ Other asexual methods

- **Budding**: in yeast (activity: yeast in sugar solution under a microscope shows small round buds) and in *Hydra* (repeated cell division at one site forms a bud that grows and detaches; many buds may appear together).
- **Spore formation**: moulds like *Rhizopus* and *Aspergillus* produce millions of light, usually single-celled spores in sacs or on vesicles at hyphal tips; spores float in air and germinate in moisture and nutrients (so bread grows mould in a warm moist chamber, 25–35 °C; refrigeration slows this).
- **Louis Pasteur** proved that life comes from pre-existing life, ending spontaneous generation, and supported germ theory and sterilisation.
- Fungi are useful: degrade wastes and pollutants, remove heavy metals, give antibiotics (penicillin).

> **IMPORTANT**
> In all asexual methods the central process is **mitosis**, so the offspring are **genetically identical clones** of the parent. Uniform clones are however more vulnerable to the same disease.

12.2 Sexual Reproduction

▸ Meiosis and variation

- Each species has a fixed chromosome number. If both parents gave full sets, numbers would double each generation; **meiosis** solves this.
- Humans have 23 pairs (46) chromosomes. In meiosis a **diploid** cell becomes **haploid** gametes (23 chromosomes): sperm and egg in animals; pollen (male gametes) and ovule (egg) in plants.
- Each gamete gets one chromosome from every pair. Bead activity: with just 3 pairs of characters, 2^3 = **8 combinations** are possible; with 23 pairs the number is huge, so children differ from parents and siblings.
- Variation helps some individuals survive changes (e.g. tolerance of low oxygen at altitude, adult milk digestion) and drives evolution.

▸ Flower — the reproductive organ of angiosperms

Whorl / part	Sub-parts	Function
Sepals	Green outermost whorl	Protect the bud and the flower
Petals	Coloured, often fragrant	Attract pollinators
Stamen (male)	Filament + **anther** (makes pollen grains with male gametes)	Produces male gametes
Pistil (female)	**Stigma** (tip, sticky/flat), **style** (tube), **ovary** (contains ovules; each ovule has an egg cell)	Receives pollen; produces egg

A complete flower has all four whorls. Pea-plant activity: wrapping or removing stamens shows that fruit forms only when pollen reaches the stigma; fruit formed in all treatments except the bud whose stamens were removed and bagged.

▸ Pollination

DEFINITION

Pollination: transfer of pollen from anther to stigma. **Self-pollination**: to the stigma of the same flower or another flower of the same plant. **Cross-pollination**: to a flower of another plant of the same kind (increases variation).

Agent	Plant examples	Adaptations
Wind	Wheat, maize, rice	Light small pollen in huge numbers; long feathery stigma
Water	*Vallisneria*, *Hydrilla*	Water currents carry pollen
Insects	Sunflower, hibiscus, marigold	Bright colour, nectar, fragrance; large sticky or spiny pollen; sticky stigma
Birds	Coral tree, hibiscus	Visited by sunbirds, white-eyes

Strategy	Pollen per flower	Seeds formed
Wind (maize, wheat)	5,00,000 – 10,00,000	50 – 200
Insect (sunflower)	20,000 – 40,000	800 – 1,000

Insect pollination wastes less pollen (higher seeds per pollen); wind pollination compensates with very large numbers of pollen grains.

▸ Fertilisation and seed formation

Pollen lands
On a compatible stigma, a pollen grain germinates into a **pollen tube**.

Tube grows
Down the style into the ovary; the male gamete travels through it to the ovule.

Fusion
Male gamete fuses with the egg: **fertilisation**; the fertilised egg is a **zygote**, which becomes an embryo.

Fruit and seed
Ovule → seed; ovary → fruit. Seeds are dispersed by wind, water or animals and germinate in favourable conditions.

- Plant breeding uses sexual reproduction: **selective breeding**, **artificial hybridisation** (remove stamens, bag the flower, hand-pollinate with chosen pollen) and **genetic engineering** to get high-yield and disease-resistant varieties.
- P. Maheshwari, 'Father of Indian Embryology', developed in-vitro fertilisation in flowering plants.

12.3 Reproduction in Animals

Type	Where fertilisation occurs	Examples	Consequence
External fertilisation	Outside the body, in water; female releases eggs, male releases sperm over them	Fish, frogs	Many eggs (fish hundreds–thousands; frog 5,000–50,000) as many are lost; survival low
Internal fertilisation	Inside the female body	Reptiles (lizard 2–20 eggs), birds (1–15 eggs), mammals	Fewer eggs; better protection; survival moderate to high

- Fish, amphibians and insects lay many yolk-poor eggs; the **larva** hatches, feeds, and then transforms (e.g. butterfly: egg → larva → pupa → adult; frog: tadpole).
- Reptiles and birds lay eggs with enough yolk to nourish the embryo until hatching; in mammals the zygote develops inside the mother and the young are fed milk.
- Care after birth depends on how developed the young are at hatching or birth.

12.4 Reproduction in Human Beings

At reproductive maturity (puberty) organs mature and produce gametes. A sperm fusing with an egg gives a zygote → embryo → **foetus** (from about the ninth week) in the **uterus**.

System	Organs and functions
Male	**Testes** (in the **scrotum**, which keeps them slightly cooler) make sperm and a hormone; **vas deferens** carries sperm to the **urethra** (common passage for urine and sperm); **seminal vesicles and prostate** add nourishing fluid; the sperm has a head (genetic material) and a tail for swimming
Female	**Ovaries** make eggs and hormones; **oviducts (fallopian tubes)** carry the egg to the **uterus** (bag-like, where the foetus develops); it opens into the **vagina** through the **cervix**

Feature	Sperm	Egg
Size	Very small	Large
Number	Millions	Few
Stored nutrients	Absent	Present
Motility	Actively motile	Non-motile

Gametogenesis (formation of gametes in testes and ovaries) is by meiosis: gametes have 23 chromosomes; the zygote regains 46. Males form numerous motile sperm, females usually one large egg at a time.

▸ Fertilisation, implantation and the menstrual cycle

- A girl's ovaries hold millions of immature eggs from birth. From puberty usually **one mature egg is released each month** (**ovulation**) and the uterine lining thickens with blood vessels.
- If a sperm fuses with the egg in the oviduct a **zygote** forms; it divides mitotically while moving to the uterus and **implants** in the lining: the start of **pregnancy** (menstrual cycle stops).
- If the egg is not fertilised it lives about a day, then the lining breaks down and leaves with blood through the vagina: **menstruation** (3–7 days).
- The cycle repeats every 21–35 days (often 28), starting at puberty (10–14 years) and continuing until menopause (about 50 years).

Days (28-day cycle)	Event
1–5	Menstruation: lining sheds
6–14	Lining rebuilds; egg matures in ovary
About 14	Ovulation: ovary releases a mature egg
15–28	Lining thickens; if no fertilisation, it starts breaking down around day 28

> **REMEMBER**
>
> **Sex determination**: females have XX, males XY. The mother always gives X; the **father's gamete (X or Y)** decides the baby's sex.

▸ Pregnancy, childbirth and care

- Pregnancy lasts about **nine months** in three **trimesters**: 1st — embryo forms, major organs start; 2nd — foetus grows, movements felt; 3rd — rapid growth for life outside. Strong uterine contractions push the baby out; doctors may help when a normal birth is unsafe.
- Newborn care: **breastfeeding** (complete nutrition, protects from disease), warmth, vaccination, gentle handling.
- Mother needs a balanced diet, check-ups, rest, emotional support, and should avoid smoking, alcohol and medicines not advised by a doctor. Post-partum depression is treatable; **ASHA workers** (over 10 lakh) support maternal care, safe deliveries and immunisation.
- **IVF**: egg and sperm combined in a laboratory dish and the embryo placed in the uterus ('test tube baby'); India's first, Kanupriya Agarwal (Durga), was born in 1978 through work of Subhash Mukhopadhyay.
- **Menstrual hygiene**: use pads/tampons/cups, change pads every 4–6 hours, wash with water, wash hands with soap before and after, dispose pads wrapped (not flushed), dry reusable pads fully. Menstruation is a sign of a healthy system.

▸ Maturity, STIs and birth control

- Sexual maturity (sperm production, menstrual cycles) is gradual, but **emotional maturity** takes longer; responsible choices prevent unplanned pregnancy and infection.
- **STIs**: gonorrhoea, herpes, syphilis, genital warts, **HIV** (can lead to AIDS); some are incurable; condoms reduce transmission and prevent pregnancy.
- Birth control: **barriers** (condoms, vaginal covers); **oral pills** (hormonal; may have side effects, no protection against STIs); **IUDs** such as **Copper-T**; **surgical** blocking of vas deferens or fallopian tubes.
- Abortion (generally within the first trimester) is medical; **prenatal sex determination is illegal in India** to protect the sex ratio.
- CDRI Lucknow developed a once-a-week non-steroidal, non-hormonal oral contraceptive.

12.5 Quick Exam Questions

Likely question	Model answer
Why does asexual reproduction give identical offspring?	It involves only one parent and mitosis, which copies the chromosomes exactly.
What is vegetative propagation? Give examples and advantages.	New plants from vegetative parts (potato, ginger, Bryophyllum); gives identical plants quickly and on a large scale.
Differentiate cutting, grafting and layering.	Cutting: a piece planted; grafting: scion fitted on a rooted stock; layering: buried twig roots while attached to parent.
How do yeast and hydra reproduce?	By budding: an outgrowth forms, enlarges and separates.
Why does bread get mould?	Airborne spores settle on moist, warm bread and germinate rapidly; refrigeration slows this.
Why is meiosis necessary in sexual reproduction?	It halves the chromosome number so the zygote regains the normal number and creates variation.

Likely question	Model answer
Name the parts of a flower and their functions.	Sepals protect, petals attract, stamen produces pollen, pistil (stigma, style, ovary) receives pollen and holds ovules.
Differentiate self- and cross-pollination.	Self: same flower or plant; cross: between flowers of different plants of the same kind.
How is wind pollination adapted?	Light pollen in huge numbers and long feathery stigma.
Describe fertilisation in flowering plants.	Pollen tube grows to the ovule, male gamete fuses with the egg forming a zygote; ovule becomes seed, ovary fruit.
Differentiate external and internal fertilisation.	External: outside body in water, many eggs, low survival; internal: inside the female, fewer eggs, better survival.
What is the function of testes, ovary, uterus?	Testes make sperm; ovaries make eggs; the uterus nourishes the developing foetus.
What is menstruation and why does it occur?	Shedding of the uterine lining with blood when the egg is not fertilised.
Why does the menstrual cycle stop in pregnancy?	The implanted embryo needs the thick lining, so it is not shed.
Who determines the sex of the baby and how?	The father: X sperm gives a girl (XX), Y sperm a boy (XY).
Name two methods of birth control and one protection against STIs.	IUD (Copper-T) and oral pills; condoms protect against STIs.

★ KEY TAKEAWAYS

- **Asexual**: one parent, mitosis, clones (budding, spores, vegetative propagation, tissue culture); **sexual**: two parents, meiosis, variation.
- Flower: **sepal, petal, stamen (anther), pistil (stigma, style, ovary)**; **pollination → fertilisation → zygote**; **ovule → seed, ovary → fruit**.
- External fertilisation (fish, frog) needs many eggs; internal (reptiles, birds, mammals) gives better survival.
- Human gametes have **23** chromosomes; zygote **46**; **XX/XY** — the father decides sex.
- Menstrual cycle ≈ **28 days**; **ovulation** about day 14; menstruation lasts 3–7 days; pregnancy ≈ **9 months** (3 trimesters).
- Promote hygiene, balanced diet, and use **condoms/contraceptives** responsibly; sex determination before birth is illegal.

CHAPTER 13 · EARTH SCIENCE / BIOLOGY

Earth as a System: Energy, Matter and Life

Life on Earth runs on a constant flow of energy and matter. Treating the Earth as one system of interacting 'spheres', this chapter explains how uneven heating by the Sun drives winds and ocean currents, how water, carbon, nitrogen and oxygen cycle through living and non-living parts, and how human activity disturbs these balances.

Big Questions

- How does warming of the Arabian Sea affect the southwest monsoon?
- What happens to coastal cities if glaciers and polar ice melt faster?
- How would rising CO_2 affect ocean plankton?

13.1 Earth as a System of Spheres

Sphere	Meaning	Examples
Geosphere	Solid rocks, soil, landforms, Earth's interior	Deccan plateau, Thar desert
Hydrosphere	Liquid water: surface and groundwater	Oceans, Ganga–Brahmaputra, lakes
Cryosphere	Water in solid form	Himalayan glaciers, snow in Ladakh, polar caps
Atmosphere	Air surrounding the Earth	Gases we breathe
Biosphere	All living organisms and their habitats	Mangroves, forests, farms, plankton, coral reefs

IMPORTANT

A disturbance in one sphere changes others. Example chain: less winter snow (cryosphere) → lower lake level (hydrosphere) in summer → less grass (biosphere) for sheep. On a large scale: warmer Arabian Sea → more evaporation → variable southwest monsoon (floods in some places, drought in others) → warming melts glaciers and polar ice → sea-level rise threatens coasts and ecosystems.

13.2 Uneven Heating of the Earth

▸ Solar radiation and the electromagnetic spectrum

- The Sun is the main energy source. Its energy arrives as **electromagnetic (EM) waves** that travel through vacuum at the speed of light, **3×10^8 m s^{-1}**. (Unlike sound, EM waves need no medium.)
- The spectrum runs from high frequency / short wavelength (gamma rays, X-rays — very energetic and harmful) through UV, visible and IR, to long wavelength microwaves and radio waves.
- About **99%** of solar energy is in **UV, visible and infrared**. Gamma and X-rays are mostly filtered by the upper atmosphere; microwaves and radio waves carry little energy.

Radiation	Role
Ultraviolet (100–400 nm)	Mostly absorbed by the **ozone layer**; excess damages eyes and skin and raises cancer risk; also kills germs in water purifiers

Radiation	Role
Visible	Reaches the surface; energy for **photosynthesis**; partly warms land and water
Infrared	Warms the surface, which re-radiates heat; part of outgoing heat is trapped by **greenhouse gases** (CO_2, CH_4, water vapour)

DEFINITION

Insolation: solar radiation reaching the Earth's surface. **Solar constant**: average solar energy received per unit time per unit area perpendicular to the Sun's rays at the top of the atmosphere, about **1.4 kW m^{-2}** (1400 J s^{-1} m^{-2}). Because of absorption and scattering, maximum insolation at the surface is about **1 kW m^{-2}** on a clear day.

FORMULA

Energy = intensity × area × time. Example: 1 kW m^{-2} on 1 m^2 for 1 hour = 1000 × 1 × 3600 = **3.6×10^6 J** (≈ one unit of household electricity).

- India lies in tropical and sub-tropical latitudes, so it gets plentiful sunshine — it drives the southwest monsoon and gives huge solar-power potential. **Anna Mani** mapped insolation across India (*Solar Radiation Over India*, 1981).
- Even a fraction of the Thar desert covered with solar panels could meet India's electricity demand (estimation exercise).

▸ Surface effects: albedo and re-radiation

Albedo is the fraction of solar radiation a surface reflects (Latin for whiteness). High albedo means cool; low albedo means hot.

Surface	Albedo	Effect
Snow	0.80–0.90	Reflects most; polar regions stay cold
Ice	0.50–0.70	High reflection
Crushed rock	0.25–0.30	Moderate
Black soil, ocean water	Low	Absorb more; relatively warm

- Dark roads and clothes heat quickly; light ones stay cooler. Concrete houses re-radiate heat at night, while thick mud-and-wood walls stay cooler.
- **Urban heat island effect**: cities are warmer than the rural surroundings because steel, concrete, brick and asphalt absorb and re-radiate heat; vegetation in rural areas gives shade and transpiration cooling. It raises air-conditioning demand.

▸ Latitude and Earth's shape

Earth is a sphere with a tilted axis, so the Sun's rays strike different latitudes at different angles. At the equator the same energy is concentrated over a **small area**; at the poles it is spread over a **large area**. Hence the equator is warm all year and the poles are cold. This temperature difference drives global winds and ocean currents and, with the tilt, causes seasons and changing day length.

▸ Role of the atmosphere

Held by gravity, air is **78% nitrogen and 21% oxygen** with a little argon, CO_2, water vapour and others. It has layers:

Layer	Height (approx.)	Features
Troposphere	0–12 km (highest over equator, lowest over poles)	Nearly all weather; heated from the surface, **temperature falls with height (~6.5 °C per km)**; rising warm air drives winds and storms
Stratosphere	12–50 km	**Ozone layer** absorbs UV so **temperature rises with height**; stable, little vertical mixing, so weather stays in troposphere
Mesosphere, thermosphere, exosphere	Above 50 km; outer space begins about 100 km	Minor role in surface climate

- Two protective roles: it **partly absorbs incoming radiation** (ozone, clouds, gases) and it **traps outgoing heat** through greenhouse gases. Without it Earth would be too cold; excess CO_2 from humans strengthens the greenhouse effect causing **global warming**.
- Venus is hotter than Mercury because its atmosphere caused a runaway greenhouse effect.
- **Ozone and CFCs**: CFCs from refrigerators and aerosols caused the Antarctic ozone hole; the **Montreal Protocol** curbed CFCs and the layer is slowly recovering.
- K. R. Ramanathan studied ozone at altitude and led early monsoon forecasting.

13.3 Winds and Ocean Currents

Wind is air moving from high to low pressure; pressure differences arise from uneven heating.

Local wind	When	Mechanism
Land and sea breeze	Day / night	Land heats and cools faster than water
Valley breeze	Day	Sun-facing slopes heat faster, air rises (low pressure) and cool air moves up from the valley
Mountain breeze	Night	Slopes cool faster; cold dense air sinks down to the valley (felt in Shimla, Dehradun)

▸ Planetary winds

- Intense heating at the **equator** makes warm air rise (**equatorial low**); it moves poleward aloft, cools and sinks near **30° N and S** (**sub-tropical highs**), and returns along the surface to the equator.
- Part of the surface air moves poleward, rises near **60° N and S** where it meets cold polar air (**sub-polar lows**). At the poles (90°), cold dense air sinks (**polar highs**) and flows to the sub-polar belts.
- Earth's rotation deflects winds: to the **right in the Northern** and to the **left in the Southern Hemisphere**, so they follow curved paths.

▸ Ocean currents

- Continuous movement of large water masses. Planetary winds drag the surface water by friction; temperature and salinity differences, Earth's rotation and continents also shape them.

- Warm surface water flows from the equator toward poles; cold, dense water returns toward the equator at depth. Low-salinity water stays near the surface; **high-salinity dense water sinks**.
- Rotation produces circular **gyres**: **clockwise in the Northern**, **counter-clockwise in the Southern Hemisphere**.
- They carry heat from equator to poles (reducing temperature differences), carry nutrients for ecosystems and moderate climate. The warm **North Atlantic Drift** (extension of the **Gulf Stream**) keeps ports of north-western Europe ice-free in winter.
- IITM Pune runs coupled atmosphere–ocean–land–ice computer models to forecast the Indian monsoon.

13.4 Biogeochemical Cycles

DEFINITION

Biogeochemical cycle: the cyclic movement of matter (and energy) between the non-living (abiotic) and living (biotic) parts of the Earth, so that nutrients like carbon, nitrogen and oxygen are recycled and stay available for life.

▸ Water cycle

- Processes: evaporation, transpiration, condensation (clouds), precipitation (rain, hail, snow), run-off, infiltration and groundwater. Water dissolves minerals and carries them to the ocean.
- **Climate change effect**: warmer air holds more moisture, so heavier rain in some places (intense monsoon) and drought elsewhere; melting glaciers raise sea level (threat to Mumbai, Chennai); intense bursts increase run-off and soil erosion and reduce infiltration and groundwater recharge, hurting agriculture in dry months.
- The cycle links cryosphere, hydrosphere, atmosphere, geosphere and biosphere.

▸ Carbon cycle

- Carbon is the backbone of proteins, carbohydrates, fats and DNA (about 49% of dry weight of organisms). It moves between atmosphere (CO_2), biosphere, geosphere (carbonate rocks, fossil fuels) and hydrosphere (dissolved CO_2, shells).
- **Fast cycle (days–years)**: photosynthesis takes CO_2 into glucose; respiration, decomposition and animals' death return it.
- **Slow cycle (millions of years)**: buried dead organisms form coal, oil and gas; **burning** releases CO_2 quickly.
- Oceans absorb CO_2 as carbonate and bicarbonate ions used by phytoplankton and shell-forming organisms; about **71%** of global carbon is in oceans, only ~1% in the atmosphere.
- Burning fossil fuels and deforestation raised atmospheric CO_2 by about **35% since 1960 (315 → 420 ppm)** — the Keeling curve with seasonal sawtooth from Northern Hemisphere plant growth. Excess CO_2 → stronger greenhouse effect, warming, melting ice, sea-level rise, extreme weather, more acidic oceans.

▸ Nitrogen cycle

Nitrogen is needed for proteins and nucleic acids. The air is rich in N_2, but it is unreactive and cannot be used directly by plants and animals.

Step	What happens	Agents
Nitrogen fixation	N_2 → ammonia (NH_3) (also nitrogen oxides by lightning)	*Rhizobium* (legume root nodules), *Azotobacter* (soil)
Nitrification	Ammonia → nitrite (NO_2^-) → nitrate (NO_3^-)	*Nitrosomonas*, *Nitrobacter*
Assimilation	Plants take up nitrogen compounds; animals eat plants/animals	Plants, animals
Ammonification	Dead matter and waste → ammonia in soil	Decomposers (bacteria, fungi)
Denitrification	Nitrates → N_2 gas returned to air	*Pseudomonas*

The **Haber-Bosch process** (early 1900s) makes ammonia from atmospheric nitrogen for fertilisers ('bread from air'), enabling the Green Revolution, but it is energy intensive (1–2% of global energy) and fertiliser over-use degrades soil and water.

▸ Oxygen cycle

- Oxygen forms 21% of air and is part of carbohydrates, proteins, nucleic acids and fats; also found in metal oxides, minerals and CO_2.
- **Used** in respiration of plants and animals, in combustion of fuels, and in oxide formation; **restored** mainly by **photosynthesis** (sunlight + water + CO_2 → glucose + O_2).

Cycle	Main reservoir	Key processes	Human disturbance
Water	Oceans	Evaporation, condensation, precipitation, run-off	Warming, altered rainfall, melting ice
Carbon	Oceans (71%)	Photosynthesis, respiration, decay, burning	Fossil fuels, deforestation
Nitrogen	Atmosphere	Fixation, nitrification, assimilation, ammonification, denitrification	Fertiliser over-use, eutrophication
Oxygen	Atmosphere	Respiration and combustion vs photosynthesis	Deforestation, burning

13.5 Human Impact and Restoring Balance

- **Ocean acidification**: excess CO_2 makes seawater acidic, threatening plankton and coral reefs; warmer water also absorbs less CO_2 (weaker carbon sink).
- **Eutrophication**: excess nitrates from fertilisers wash into rivers and lakes, causing algal blooms that deplete oxygen and kill fish.
- **Deforestation**: less photosynthesis and transpiration (lower local rainfall), changed albedo, soil erosion without roots, habitat and biodiversity loss.
- **Smog and ground-level ozone** from vehicle emissions harm health (stratospheric ozone is protective).
- Positive steps: Montreal Protocol (success), solar and wind energy, planting trees, saving water, sustainable farming, reduce–reuse–recycle; Kyoto Protocol and Paris Agreement have been less successful.
- **Mission LiFE** (Lifestyle for Environment), India-led initiative (2021), promotes mindful eco-friendly living.

13.6 Quick Exam Questions

Likely question	Model answer
Name the five spheres of the Earth system with an example each.	Geosphere (rocks), hydrosphere (rivers), cryosphere (glaciers), atmosphere (air), biosphere (forests).
Why is the Earth unevenly heated?	Because it is spherical and tilted, so rays strike latitudes at different angles; the equator gets concentrated radiation and the poles spread-out radiation; surfaces also differ in albedo.
Define solar constant and insolation.	Solar constant: about 1.4 kW m^{-2} at the top of the atmosphere; insolation: solar radiation reaching the surface (max about 1 kW m^{-2}).
Calculate energy on 1 m^2 in one hour at 1 kW m^{-2}.	$E = 1000 \times 1 \times 3600 = 3.6 \times 10^6$ J.
What is albedo? How does it affect temperature?	Fraction of radiation reflected; high-albedo snow stays cold, low-albedo black soil and oceans warm up.
Why do cities become heat islands?	Concrete, asphalt and steel absorb and re-radiate heat, with less vegetation for shade and transpiration.
Why does weather occur in the troposphere?	It is heated from the surface, warm air rises and drives winds, clouds and storms; the stable stratosphere above restricts mixing.
How do greenhouse gases keep the Earth warm?	They trap infrared re-radiated from the heated surface; without them the Earth would be too cold.
Explain valley and mountain breezes.	By day slopes warm, air rises and valley air moves up; at night slopes cool and dense air flows down.
What causes planetary winds and their deflection?	Pressure belts from equator-to-pole heating; Earth's rotation deflects them right in the north and left in the south.
What are ocean currents and why are they important?	Large steady movements of ocean water; they move heat and nutrients and moderate climate (Gulf Stream keeps European ports ice-free).
Describe the steps of the nitrogen cycle.	Fixation, nitrification, assimilation, ammonification, denitrification.
Why can plants not use atmospheric N_2 directly?	It is unreactive; bacteria must convert it to ammonia, nitrite or nitrate.
Why is excess CO_2 harmful although plants need it?	It enhances the greenhouse effect, causing warming, ice melt, sea-level rise, extreme weather and acidic oceans.
What is eutrophication?	Algal blooms caused by excess fertiliser nitrates that use up oxygen and kill fish.
Mention two effects of deforestation.	Reduced rainfall and photosynthesis, soil erosion and loss of biodiversity.

★ KEY TAKEAWAYS

- Earth is an interacting system of **geosphere, hydrosphere, cryosphere, atmosphere, biosphere**.
- Solar radiation (UV, visible, IR) heats the Earth **unevenly** — by latitude, albedo and atmosphere; **solar constant 1.4 kW m^{-2}**.
- **Troposphere** (0–12 km) has the weather; the **ozone layer** in the stratosphere absorbs UV; **greenhouse gases** trap heat.
- Uneven heating drives **local winds, planetary winds and ocean currents (gyres)**.
- **Water, carbon, nitrogen and oxygen cycles** recycle matter between living and non-living parts.
- Fossil fuels, fertilisers and deforestation disturb cycles; **cooperation (Montreal Protocol) and sustainable living (Mission LiFE)** help restore balance.